

Transmission Control Protocol (TCP): Connection Phases, Wrap-around Time, and Structural Mechanics

Summary

TCP is a **connection-oriented, reliable, full-duplex transport protocol** that ensures ordered delivery, error control, and flow management over an unreliable IP network. This document synthesizes key aspects of **wrap-around time (WAT)**, **sequence number management**, **connection phases**, and **header mechanics**.

Key takeaways:

- TCP connections are **virtual**, with buffers pre-allocated but segments potentially traversing different paths.
- **Sequence numbers** are 32-bit; control flags (SYN/FIN) consume one sequence number each, pure ACKs consume none.
- **Wrap-around time (WAT)** is bandwidth-dependent; high-speed networks require Time Stamps to avoid ambiguity.
- Connections follow three phases: **Establishment**, **Data Transfer**, and **Termination**, with sequence number tracking throughout.

1. Wrap-around Time (WAT) Analysis

TCP assigns a **unique sequence number to every byte**. WAT measures how long it takes to exhaust the 32-bit sequence number space.

1.1 Calculation

$$\text{WAT} = \frac{\text{Total Sequence Numbers}}{\text{Bandwidth (Bytes/sec)}}$$

- Convert **bandwidth** from Mbps to **Bytes/sec**:

$$\text{Bps} = \frac{\text{Mbps}}{8} \times 10^6$$

- **Total Sequence Numbers:** 2^{32} by default ($\approx 4.29 \times 10^9$ bytes).

1.2 Minimum Bits Required

To prevent reuse within the Maximum Segment Lifetime (MSL):

$$\lceil \log_2(\text{Life Time} \times \text{Bandwidth in Bytes/sec}) \rceil$$

Bandwidth	Sequence Field	Life Time	Minimum Bits / WAT
200 Mbps	28-bit	60s	~10.7 s
1 Gbps	32-bit	180s	~34.35 s
1 Gbps	Variable	60s	33 bits required
100 Mbps	Variable	120s	31 bits required

2. Structural Characteristics of TCP Connections

- **Virtual Connection:** TCP is logical; segments may traverse different IP paths, arrive out of order, or be lost.
- **Full Duplex:** Both endpoints can send and receive simultaneously.
- **Flow and Error Control:** TCP ensures data integrity end-to-end using sliding windows (GBN/Selective Repeat).

2.1 Windowing

TCP manages **four windows** per connection:

1. Client Sender Window
 2. Server Receiver Window
 3. Server Sender Window
 4. Client Receiver Window
- **Maximum Segment Size (MSS):** Maximum data in one segment (excludes IP/TCP headers). Example: Ethernet MTU = 1500 bytes → MSS ≈ 1460 bytes.

3. Three Phases of a TCP Connection

3.1 Phase I: Connection Establishment (Three-Way Handshake)

Step	Segment Type	Flags	Sequence Number Consumption
1	Client → Server	SYN=1	1
2	Server → Client	SYN=1, ACK=1	1
3	Client → Server	ACK=1	0 (pure ACK)

- Ensures ISNs and buffer allocation are synchronized.

3.2 Phase II: Data Transfer

- **Data Bytes:** Consume 1 sequence number per byte.
- **ACKs:** Represent next expected byte, may be piggybacked on outgoing segments.
- Example: Receiver gets bytes 1001–1100 → sends ACK = 1101.

3.3 Phase III: Connection Termination (Four-Way Handshake)

Step	Segment Type	Flags	Sequence Number Consumption
1	Client → Server	FIN=1	1
2	Server → Client	ACK=1	0
3	Server → Client	FIN=1	1
4	Client → Server	ACK=1	0

- Can be compressed to three steps if both ends close simultaneously.

4. TCP Control Flags and Sequence Number Behavior

Flag / Data	Sequence Number Consumption	Usage Phase
SYN	1	Establishment
FIN	1	Termination
Pure ACK	0	Any
Data Byte	1 per byte	Data Transfer
SYN+ACK	1	Establishment

Flag Combinations:

- Request: SYN=1, ACK=0
- Reply: SYN=1, ACK=1
- Data / ACK: SYN=0, ACK=1

5. MSS and Options

- MSS is **negotiated via Options** during handshake.
- Informs sender of **maximum data per segment** and receiver buffer limits.
- Example: MSS=1460 bytes, Window=14600 bytes → sender may send 10 segments before waiting for ACK.